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Instrument panel systems and methods for aircraft cockpits

Abstract

A replacement instrument panel assembly for an aircraft having an airframe and an upper skin, the replacement instrument panel assembly has a support system, a base panel assembly, a secondary panel assembly, and a skin extension assembly. The support system is adapted to be secured to the airframe. The base panel assembly is supported by the support system and comprises a base panel member defining at least one main base panel opening. The secondary panel assembly comprises at least one secondary panel member and at least one secondary panel screw. Each secondary panel member defines at least one secondary panel opening and is secured to the base panel by the at least one secondary panel screw. The skin extension assembly comprises a skin extension member supported by the support system and adapted to be secured to a portion of the upper skin.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5995290	12/1998	Noble	348/115	G02B 27/01
6255964	12/2000	Steele, Jr.	340/975	G01C 23/00
7420476	12/2007	Stiffler	701/4	H04L 67/12
2004/0200923	12/2003	Lee	244/1R	B64D 43/00
2017/0363925	12/2016	Kostrzewski	N/A	G06F 1/1637

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2023043604	12/2022	WO	N/A

OTHER PUBLICATIONS

Gibbs, Elizabeth; Old Plane, New Panel Innovative Panel Design Enables Avionics Overhaul; Cessna Owner; Oct. 2021; 4 pages; USA. cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) This application, U.S. patent application Ser. No. 18/467,477 filed Sep. 14, 2023, claims benefit of U.S. Provisional Application Ser. No. 63/375,672 filed Sep. 14, 2022, now expired, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention relates to panel systems and methods for supporting instruments in the cockpit of an aircraft and, in particular, to instrument panel systems and methods configured to allow existing aircraft to be retrofit with new or modern avionics systems.

BACKGROUND

(2) Well-maintained small private aircraft may be in service indefinitely, and many small private aircraft currently in service have been in service since 1960's or earlier.

(3) The present invention is of particular significance when applied to certain models of Cessna aircraft manufactured from approximately 1962 to approximately 1974, and that application of the present invention will be described herein in detail (hereinafter "legacy aircraft"). The principles of the present invention may, however, be applied to other aircraft, and the present invention may have applicability to aircraft other than the legacy aircraft depicted and described herein.

(4) Avionics systems have advanced significantly in the last 25 years, and many legacy aircraft

were originally manufactured without the advanced avionics systems currently available or even knowledge of the form factor of such advanced avionics systems. Accordingly, the original instrument panel assemblies of existing legacy aircraft cannot easily be modified to accept modern avionics systems without modifications that may adversely affect the structural integrity of the aircraft.

(5) The need thus exists for improved systems and methods for retrofitting existing legacy aircraft with adversely affecting the structural integrity of the aircraft.

SUMMARY

(6) The present invention may be embodied as a replacement instrument panel assembly for an aircraft having an airframe and an upper skin. The replacement instrument panel assembly comprises a support system, a base panel assembly, a secondary panel assembly, and a skin extension assembly. The support system adapted to be secured to the airframe. The base panel assembly is supported by the support system and comprises a base panel member defining at least one main base panel opening. The secondary panel assembly comprises at least one secondary panel member and at least one secondary panel screw. Each secondary panel member defines at least one secondary panel opening and is secured to the base panel by the at least one secondary panel screw. The skin extension assembly comprises a skin extension member supported by the support system and adapted to be secured to a portion of the upper skin.

(7) The present invention may also be embodied as a method of installing a replacement instrument panel assembly in an aircraft having an airframe and an upper skin, the method comprising the following steps. At least a portion of the upper skin is removed to obtain a remaining portion of the upper skin. A support system is secured to the airframe. A base panel assembly comprising a base panel member defining at least one main base panel opening is provided. The base panel assembly is supported from the support system in a desired position relative to the airframe. At least one secondary panel assembly comprising at least one secondary panel member and at least one secondary panel screw is provided. Each secondary panel member defines at least one secondary panel opening. The at least one secondary panel member is secured to the base panel using the at least one secondary panel screw. A skin extension member is provided. At least a portion of the skin extension member is supported from the support system. At least a portion of the skin extension member is secured to the remaining portion of the upper skin.

(8) The present invention may also be embodied as a replacement instrument panel assembly for an aircraft having an airframe and an upper skin, the replacement instrument panel assembly comprising a support system, a base panel assembly, a secondary panel assembly, and a skin extension assembly. The support system is adapted to be secured to the airframe and comprises a perimeter support assembly, an inner support assembly, a yoke support adapter assembly, and a plurality of support system rivets. The base panel assembly supported by the support system and comprises a base panel member defining at least one main base panel opening. The secondary panel assembly comprises at least one secondary panel member and at least one secondary panel screw. Each secondary panel member defines at least one secondary panel opening and is secured to the base panel by the at least one secondary panel screw. The skin extension assembly comprises a skin extension member supported by the support system and adapted to be secured to a portion of the upper skin by at least one skin extension rivet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side elevation view of a legacy aircraft of the type that may be retrofit with a replacement instrument panel system of the present invention;

(2) FIG. 2 is a perspective view depicting a cockpit of a legacy aircraft and an original instrument

panel assembly supported within that cockpit;

(3) FIG. 3 is a front elevation view depicting the original instrument panel supported within the cockpit of a legacy aircraft;

(4) FIG. 4 is a perspective view depicting the cockpit of a legacy aircraft and a portion of a first example replacement instrument panel assembly of the present invention supported within that cockpit;

(5) FIG. 5 is a front elevation view depicting the first example replacement instrument panel system supported within the cockpit area of a legacy aircraft;

(6) FIG. 6 is a side elevation view depicting the first example replacement instrument panel system supported within the cockpit area of a legacy aircraft;

(7) FIG. 7 is a partially exploded perspective view depicting the first example replacement instrument panel system and relevant structure of the airframe of a legacy aircraft;

(8) FIG. 8 is a section view taken along lines 8-8 in FIG. 5;

(9) FIG. 9 is a front elevation view of an example base panel member of the first example replacement panel assembly;

(10) FIG. 10 is a front elevation view depicting a first example set of secondary panel assemblies that may be combined with the first example base panel member to form a first example replacement panel assembly of the present invention;

(11) FIG. 11 is a front elevation view depicting the combination of the first example set of secondary panel assemblies and the first example base panel member to form the first example replacement panel assembly;

(12) FIG. 12 is a front elevation view depicting a second example set of secondary panel assemblies that may be combined with the first example base panel member to form a second example replacement panel assembly of the present invention;

(13) FIG. 13 is a front elevation view depicting the combination of the second example set of secondary panel assemblies and the first example base panel member to form the second example replacement panel assembly;

(14) FIG. 14 is a front elevation view depicting a third example set of secondary panel assemblies that may be combined with the first example base panel member to form a third example replacement panel assembly of the present invention;

(15) FIG. 15 is a front elevation view depicting the combination of the third example set of secondary panel assemblies and the first example base panel member to form the third example replacement panel assembly; and

(16) FIG. 16 is a front elevation of a portion of one of the example secondary panels of the third example set of secondary panels illustrating first, second, and third areas of that example secondary panel.

DETAILED DESCRIPTION

(17) Referring initially to FIG. 1 of the drawing, depicted at 20 therein is an example aircraft in connection with which the present invention may be used. The example aircraft 20 is or may be conventional and will be described herein only to that extent helpful to a complete understanding of the present invention.

(18) FIG. 1 illustrates that the example aircraft 20 defines a cockpit area 22, and FIGS. 2 and 3 illustrate an example original instrument panel assembly 24 supported within that cockpit area 22. FIGS. 2 and 3 further illustrates a portion of an example airframe assembly 30 of the example aircraft 20. The example airframe assembly 30 supports an outer skin 32, and a cross-hatched area of FIG. 1 illustrates an upper skin portion 34 of the outer skin 32. The upper skin portion 34 extends from a firewall 36 to the cockpit area 22.

(19) FIGS. 2 and 3 illustrate that the example airframe assembly 30 comprises first, second, third, and fourth airframe portions 40, 42, 44, and 46 and that the airframe assembly 30 further supports a center console support assembly 50 and a yoke support assembly 52. FIGS. 2 and 3 further

illustrate that the example original instrument panel **24** defines a first yoke opening **60**, a second yoke opening **62**, and one or more instrument openings **64**.

(20) The location and structure of the example airframe assembly **30**, center console support assembly **50**, and yoke assembly **52** are not substantially changed or modified by the systems and methods of the present invention. Further, the locations of the first and second yoke openings **60** and **62** and the instrument opening(s) **64** are not affected or changed the systems and methods of the present invention.

(21) Turning now to FIGS. **3-11** of the drawing, depicted therein is a first example replacement instrument panel system **120** comprising a first example replacement panel assembly **122**. The first example replacement panel assembly **122** comprises a base panel assembly **130**, a first example secondary panel assembly **132**, a skin extension assembly **134**, and a support system **136**.

(22) FIGS. **4** and **9** illustrate that the example base panel assembly **130** comprises a first example base panel member **140** and a plurality of base panel rivets **142**. FIGS. **4** and **9** further illustrate that the first example base panel member **140** defines one or more main base panel openings **150**, one or more ancillary base opening(s) **152**, a plurality of control openings **154**, and one or more yoke openings **156**. Other openings are typically formed in the first example base panel member **140** to facilitate assembly of the first example replacement panel assembly **122** as will be described in further detail below. The first example base panel member **140** defines first, second, third, fourth, fifth, and sixth primary base openings **150a**, **150b**, **150c**, **150d**, and **150e**.

(23) FIG. **9** further illustrates that first example panel member **140** further defines one or more yoke portions **158**. In the first example base panel member **140**, a first yoke opening **156a** is formed in a first yoke portion **158b**, and a second yoke opening **156b** is formed in a second yoke portion **158b**.

(24) FIGS. **10** and **11** illustrate that first example secondary panel assembly **132** comprises at least one, and typically a plurality of, secondary panel member(s) **160** and a plurality of secondary panel assembly screws **162**. Each of the secondary panel member(s) **160** defines at least one, and typically a plurality of, secondary panel opening(s) **170**. As perhaps best shown in FIGS. **10** and **11**, the first example secondary panel assembly **132** comprises first, second, third, fourth, and fifth example secondary panel members **160a**, **160b**, **160c**, **160d**, and **160e**. Each of the example secondary panel members **160a**, **160b**, **160c**, **160d**, and **160e** in turn defines a plurality of the secondary panel openings **170**.

(25) FIGS. **4**, **6**, and **7** illustrate that the example skin extension assembly **134** comprises a skin extension member **220** and a plurality of skin extension rivets **222**.

(26) FIG. **7** perhaps best illustrates that the example support system **136** comprises a perimeter support assembly **230**, an inner support assembly **232**, a yoke support assembly **324**, and a plurality of support system rivets **236**. The example perimeter support assembly **230** comprises a center upper support member **240**, a first side upper support member **242**, a second side upper support member **244**, a reinforcing member **246**, a lower support member **280**, a first side lower support member **282**, and a second side lower support member **284**.

(27) FIG. **8** illustrates that the example center upper support member **240** defines a profile having a first center upper support member portion **250**, a second center upper support member portion **252**, and a third center support member portion **254**.

(28) Each of the first and second side upper support members **242** and **244** comprises a transition portion **260** and a connecting portion **262**. The transition portions **260** are adapted to be connected to the first center upper support member **240**, and the connecting portions **262** are adapted to be secured to the first and second airframe portions **40** and **42**.

(29) The example reinforcing member **246** defines a lateral portion **270** and first and second angled portions **272** and **274**. When assembled, the example lateral portion **270** is aligned with and reinforces the first center upper support member **240**, while the first and second angled portions **272** and **274** align with and reinforce the transition portions **260** of the first and second upper

support members **242** and **244**, respectively. The example reinforcing member **246** thus reinforces the connections between the center upper support member **240** and the first and second side upper support members **242** and **244** when assembled.

(30) The example lower support member **280** defines a lower support member main portion **290** and one or more lower support member edge portions **292**. The example lower support member **280** defines first, second, third, and fourth lower support member edge portions **292a**, **292b**, **292c**, and **292d** that extend at approximately a right angle from the lower support member main portion **290**. The example lower support member **280** is generally in the shape of a five sided rectangular box and is adapted to engage first and second lower support members **282** and **284** and to engage and reinforce the base panel member **140**. At least one lower support member control opening **294** is formed in the lower support member main portion **290**.

(31) Each of the first and second side lower support members **282** and **284** defines a lower portion **320** and an upper portion **322**. The lower portion **320** is sized and dimensioned to engage the third and fourth airframe portions **44** and **46**, and the upper portions **322** are adapted to engage the first and second side upper support members **242** and **244**, respectively.

(32) The example inner support assembly **232** comprises a first side mounting member **330**, a second side mounting member **332**, and a plurality of mounting member rivets **334**. The inner support assembly **232** is or may be conventional and is designed to support a radio unit (not shown).

(33) The example yoke support adapter assembly **234** comprises a yoke support member **340** and a plurality of yoke support rivets **342**. The example yoke support member **340** is generally in the shape of a five sided rectangular box and is adapted to engage the example yoke support assembly **52**.

(34) To assemble the first example replacement instrument panel assembly **122**, the original instrument panel assembly **24** is removed from the aircraft **20**, leaving the first, second, third, and fourth airframe portions **40**, **42**, **44**, and **46**, the center console support assembly **50**, and the yoke support assembly **52**. The upper skin portion **34** is modified by cutting back the skin portion along a line adjacent to the firewall **36**, leaving a skin portion attachment portion **350**.

(35) The first example replacement panel assembly **122** is next formed by using the support system rivets **236**. In particular, the center upper support member **240**, the first and second side upper support members **242** and **244**, the reinforcing member **246**, the lower support member **280**, and first and second side lower support members **282** and **284** are riveted to the base panel member **140**.

(36) The perimeter support assembly **230** is secured to the airframe assembly **30** using the support system rivets **236**. In particular, the connecting portions **262** of the first and second side support members **242** and **244** and the first and second lower support members **282** and **284** are secured by the support system rivets **236** to the airframe assembly **30** and, in particular, to the first, second, third, and/or fourth airframe portions **40**, **42**, **44**, and **46**.

(37) With the perimeter support assembly **230** secured to the airframe assembly **30**, the base panel assembly **130** is supported relative to the airframe assembly **30**. In addition, the base panel assembly **130** may further be riveted to the center console assembly **50** and to the yoke support assembly **52** through the yoke support adapter assembly **234**. Further, the skin extension member **220** is secured to the perimeter support assembly **230** by the support system rivets **236** and to the skin portion attachment portion **350** by skin extension rivets **222** as perhaps shown in FIGS. **4** and **6**. With the base panel assembly **130** supported by the airframe assembly **30**, the base panel **130** is in a desired spatial position and orientation within the cockpit area **22**.

(38) At this point, the secondary panel member(s) **160** are secured to the base panel assembly **130** to form the replacement instrument panel assembly **122** such that at least one of the secondary panel members **160** is secured to the base panel member **140** by the secondary panel screws **162** such that desired secondary panel openings **170** are arranged within at least one of the primary base

openings **150**. In the example depicted in FIGS. **9-11**, the first, second, third, fourth, and fifth example secondary panel members **160a**, **160b**, **160c**, **160d**, and **160e** are secured to the base panel member **140** by the secondary panel screws **162** such that desired secondary panel openings **170** are arranged within at least one of the first, second, third, fourth, and fifth primary base openings **150a**, **150b**, **150c**, **150d**, and **150e** defined by the base panel member **140**. The example secondary panel openings **170** are associated with a typical Dynon Electronic Flight Display (EFD) System configuration. Instruments such as display panels, gauges, switches, and the like (not shown) of a Dynon EFD System may thus be supported by the first example replacement panel assembly **122**.

(39) Turning now to FIGS. **12** and **13** of the drawing, depicted therein is a second example panel replacement assembly **420** comprising a second example secondary panel assembly **422** that may be connected to the first example base panel member **140** as part of the first example replacement panel system **120**. FIGS. **12** and **13** illustrate that second example secondary panel assembly **422** comprises at least one, and typically a plurality of, secondary panel member(s) **430** and a plurality of secondary panel assembly screws **432**. Each of the secondary panel member(s) **430** defines at least one, and typically a plurality of, secondary panel opening(s) **440**. FIGS. **12** and **13** further illustrate that the second example secondary panel assembly **422** comprises first, second, third, fourth, and fifth example secondary panel members **430a**, **430b**, **430c**, **430d**, and **430e**. Each of the example secondary panel members **430a**, **430b**, **430c**, **430d**, and **430e** in turn defines a plurality of the secondary panel openings **440**. The example secondary panel openings **440** are associated with a typical Garmin EFD System configuration. Instruments such as display panels, gauges, switches, and the like (not shown) of a Garmin EFD System may thus be supported by the second example replacement panel assembly **420**.

(40) FIGS. **14** and **15** of the drawing illustrate a third example panel replacement assembly **450** comprising a third example secondary panel assembly **452** that may be connected to the first example base panel member **140** as part of the first example replacement panel system **120**. FIGS. **14** and **15** illustrate that third example secondary panel assembly **452** comprises at least one, and typically a plurality of, secondary panel member(s) **460** and a plurality of secondary panel assembly screws **462**. Each of the secondary panel member(s) **460** defines at least one, and typically a plurality of, secondary panel opening(s) **470**. FIGS. **14** and **15** further illustrate that the third example secondary panel assembly **452** comprises first, second, third, fourth, and fifth example secondary panel members **460a**, **460b**, **460c**, **460d**, and **460e**. Each of the example secondary panel members **460a**, **460b**, **460c**, **460d**, and **460e** in turn defines a plurality of the secondary panel openings **470**. The example secondary panel openings **470** are associated with a typical Traditional "T" (Pilot's Six Pack) configuration with standard $3\frac{1}{8}$ and $2\frac{1}{4}$ analog gauges. Instruments such as display panels, gauges, switches, and the like (not shown) of a Pilot's Six Pack System may thus be supported by the third example replacement panel assembly **450**.

(41) The first example replacement instrument panel system **120** may thus include the base panel assembly **130** in combination with one of the first example secondary panel assembly **132**, the second example secondary panel assembly **422**, or the third example secondary panel assembly **452** depending on the configuration of the EFD System desired by the owner or operator of the aircraft **20**.

(42) Turning now to FIG. **16** of the drawing, depicted therein is a representation of the first example secondary panel member **460a** of the third example secondary panel assembly **452**. Broken lines in FIG. **16** illustrate that the first example secondary panel member **460a** may be divided into first, second, and third regions **480**, **482**, and **484**. The example first region **480** extends around a perimeter of the first example secondary panel member **460a**. The example second and third regions **482** and **484** lie within the boundaries of the first region **480**. The example first region **480** generally overlaps the surface area of the first example base panel member **140**, while the example second and third regions lie within the first primary base opening **150a** defined by the example panel member **140**. The secondary panel screws **162** extend through the first

example secondary panel member **460a** and into the first example base panel member **140** within the first region **480**.

(43) The example secondary panel opening(s) **470** defined by the first example secondary panel member **460a** are all arranged within the second region **482** so that instruments mounted within these secondary panel openings **470** are not obstructed by the first example base panel member **140**.

(44) One or more additional secondary panel openings **470** may be formed within the third region **484**, but these additional secondary panel openings **470** are limited in size and number to maintain the structural integrity supplied by the first example secondary panel member **460a** to the third example replacement panel assembly **450**. Accordingly, a limited number of control elements, such as headphone jacks, toggle switches, and/or indicator lights, that require small openings for mounting may be arranged in the third region **484**.

(45) The secondary panel members **160** and **430** may similarly define first, second, and third regions like the first, second, and third regions **480**, **482**, and **484** defined by the secondary panel member **460a** as described above.

Claims

1. A replacement instrument panel assembly for an aircraft having an airframe and an upper skin, the replacement instrument panel assembly comprising: a support system adapted to be secured to the airframe; a base panel assembly supported by the support system, the base panel assembly comprising a base panel member defining at least one main base panel opening; a secondary panel assembly comprising at least one secondary panel member and at least one secondary panel screw, where each secondary panel member defines at least one secondary panel opening, and is secured to the base panel by the at least one secondary panel screw; and a skin extension assembly comprising a skin extension member supported by the support system and adapted to be secured to a portion of the upper skin.
2. A replacement instrument panel assembly as recited in claim 1, in which the support system comprises a perimeter support assembly, an inner support assembly, a yoke support adapter assembly, and a plurality of support system rivets.
3. A replacement instrument panel assembly as recited in claim 2, in which the perimeter support assembly comprises a center upper support member, a first side upper support member, a second side upper support member, a reinforcing member, a lower support member, a first side lower support member, and a second side lower support member.
4. A replacement instrument panel assembly as recited in claim 2, in which the inner support assembly comprises a first side mounting member, a second side mounting member, and a plurality of mounting member rivets.
5. A replacement instrument panel assembly as recited in claim 2, in which the yoke support adapter assembly comprises a yoke support member and a plurality of yoke support rivets.
6. A replacement instrument panel assembly as recited in claim 1, in which the secondary panel assembly comprises a plurality of secondary panel members.
7. A replacement instrument panel assembly as recited in claim 1, in which the skin extension assembly comprises at least one skin extension rivet.
8. A method of installing a replacement instrument panel assembly in an aircraft having an airframe and an upper skin, the method comprising the steps of: removing at least a portion of the upper skin to obtain a remaining portion of the upper skin; securing a support system to the airframe; providing a base panel assembly comprising a base panel member defining at least one main base panel opening; supporting the base panel assembly from the support system in a desired position relative to the airframe; providing a secondary panel assembly comprising at least one secondary panel member and at least one secondary panel screw, where each secondary panel member defines

at least one secondary panel opening; securing the at least one secondary panel member to the base panel using the at least one secondary panel screw; and providing a skin extension member; supporting at least a portion of the skin extension member from the support system; and securing at least a portion of the skin extension member to the remaining portion of the upper skin.

9. A method as recited in claim 8, in which the support system comprises a perimeter support assembly, an inner support assembly, a yoke support adapter assembly, and a plurality of support system rivets.

10. A method as recited in claim 9, in which the perimeter support assembly comprises a center upper support member, a first side upper support member, a second side upper support member, a reinforcing member, a lower support member, a first side lower support member, and a second side lower support member.

11. A method as recited in claim 9, in which the inner support assembly comprises a first side mounting member, a second side mounting member, and a plurality of mounting member rivets.

12. A method as recited in claim 9, in which the yoke support adapter assembly comprises a yoke support member and a plurality of yoke support rivets.

13. A method as recited in claim 8, in which the secondary panel assembly comprises a plurality of secondary panel members.

14. A method as recited in claim 8, in which the skin extension assembly comprises a skin extension member and a plurality of skin extension rivets.

15. A replacement instrument panel assembly for an aircraft having an airframe and an upper skin, the replacement instrument panel assembly comprising: a support system adapted to be secured to the airframe, the support system comprises a perimeter support assembly, an inner support assembly, a yoke support adapter assembly, and a plurality of support system rivets; a base panel assembly supported by the support system, the base panel assembly comprising a base panel member defining at least one main base panel opening; a secondary panel assembly comprising at least one secondary panel member and at least one secondary panel screw, where each secondary panel member defines at least one secondary panel opening, and is secured to the base panel by the at least one secondary panel screw; and a skin extension assembly comprising a skin extension member supported by the support system and adapted to be secured to a portion of the upper skin by at least one skin extension rivet.

16. A replacement instrument panel assembly as recited in claim 15, in which the perimeter support assembly comprises a center upper support member, a first side upper support member, a second side upper support member, a reinforcing member, a lower support member, a first side lower support member, and a second side lower support member.

17. A replacement instrument panel assembly as recited in claim 15, in which the inner support assembly comprises a first side mounting member, a second side mounting member, and a plurality of mounting member rivets.

18. A replacement instrument panel assembly as recited in claim 15, in which the yoke support adapter assembly comprises a yoke support member and a plurality of yoke support rivets.

19. A replacement instrument panel assembly as recited in claim 15, in which the secondary panel assembly comprises a plurality of secondary panel members.

20. A replacement instrument panel assembly as recited in claim 15, in which the skin extension assembly comprises a plurality of skin extension rivets.
